

# PHASE 2 – REQUIREMENT ANALYSIS PHASE

**Project Title:** FitBuddy – AI Fitness Plan Generator using Gemini Models

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## 1. Purpose of Requirement Analysis

This phase identifies the functional, technical and user requirements needed to develop FitBuddy. The requirements will guide the design and development of the AI-based fitness plan generator.

## 2. User Requirements

- The user should be able to enter basic fitness-related information.
- The user should be able to select or describe a fitness goal.
- The user should be able to provide activity level and available workout days.
- The user should be able to submit the information and receive an AI-generated plan.
- The generated plan should be presented in a clear and readable format.

## 3. Functional Requirements

| Requirement      | Description                                                                           |
|------------------|---------------------------------------------------------------------------------------|
| Input Collection | Collect relevant user information such as age, goal, activity level and workout days. |
| AI Integration   | Send the required information to a Gemini model for processing.                       |
| Plan Generation  | Generate a structured fitness plan based on the supplied information.                 |
| Output Display   | Display the generated response in an understandable format.                           |
| Basic Validation | Check that required inputs are provided before generating a plan.                     |

## 4. Non-Functional Requirements

- Usability: The interface should be simple and easy to understand.
- Performance: The system should generate the response within a reasonable time, subject to the AI service and network.
- Reliability: The application should handle normal user inputs consistently.
- Maintainability: The project should be organised so that code and documentation can be updated easily.
- Security: API credentials and other private configuration details should not be exposed in public source code.

## 5. Software Requirements

- Web browser or suitable development environment
- Programming language/framework selected for the application
- Gemini API / Google AI Studio access as required by the project
- GitHub account for version control and project submission
- Internet connection for AI service access

## 6. Hardware Requirements

- Computer or laptop

- Keyboard and mouse/touchpad
- Stable internet connection
- Mobile device may be used for testing the user interface if applicable

## 7. Proposed Input Data

| Input                  | Example / Purpose                                                    |
|------------------------|----------------------------------------------------------------------|
| Age                    | Used as basic user context.                                          |
| Fitness Goal           | For example: general fitness, strength, weight management, etc.      |
| Activity Level         | Beginner, intermediate or similar user-selected level.               |
| Workout Days           | Number of days available for exercise.                               |
| Additional Preferences | Optional information such as available time or exercise preferences. |

## 8. Expected System Output

The system is expected to generate a structured fitness plan using the user's inputs. The output may include a suggested workout schedule, exercise recommendations and general fitness guidance appropriate to the information provided.

## 9. Requirement Analysis Conclusion

The identified requirements provide a basic foundation for designing and developing FitBuddy. The next phase will convert these requirements into the system design, workflow and interface structure.